RED & WHITE SINCE2008
SKILL EDUCATION
SHAPING SKILLS FOR SCALING HIGHER...!!!

👩‍🎓 **Exam Type:** Practical
⏳ **Duration:** 6 Hours

Title: "Customer Purchase Behavior Analyzer"

🎯 Objective:

To design a Python-based **Data Preprocessing and Feature Engineering pipeline** that analyzes customer purchase behavior using multi-format data sources and prepares a clean dataset for downstream analytics or machine learning tasks.

🧩 Project Definition:

You are provided with three datasets:

1. `users.csv` – Contains customer demographic and registration details.
2. `sales.json` – Contains daily sales transactions, with details like customer ID, product ID, purchase amount, and payment type.
3. `inventory.sql` – Contains product catalog data stored in SQL format.

Your task is to build a Python notebook or script that performs **end-to-end data preprocessing and feature engineering** using the following steps:

🪜 Stepwise Tasks to be Performed:

1. Data Understanding & Loading

- Import data from all three sources (`CSV`, `JSON`, and `SQL`).
- Display top 5 records and data info summary.
- Identify data types, missing values, and inconsistent records.

2. Data Cleaning

- Handle missing numerical data using **SimpleImputer (mean strategy)**.
- Handle missing categorical data using **most frequent imputation**.
- Apply **KNN Imputer** on multivariate data (optional enhancement).
- Fix invalid or inconsistent entries (e.g., wrong date formats, negative prices, etc.).

3. Outlier Handling

- Detect and remove outliers using **Z-score** and **IQR method**.
- Compare both techniques and decide which is more suitable for this dataset.
- Apply **Winsorization** for columns where removal isn't feasible.

4. Data Transformation

- Convert date columns into separate **day, month, year** features.
- Encode categorical variables using:
    - Label Encoding for binary columns.
    - One-Hot Encoding for nominal columns.
    - Ordinal Encoding for ordered categorical variables.
- Apply **binning** (e.g., segment customers into spending groups: Low, Medium, High).
- Apply **log and square root transformations** to normalize skewed data.

5. Feature Scaling

- Use **StandardScaler** and **MinMaxScaler** to scale numerical features.
- Compare the effect of scaling using summary statistics.

6. Feature Construction

- Create new features such as:
    - Average monthly spend per customer.
    - Frequency of purchase.
    - Days since last purchase.
    - Category-wise total expenditure.

7. Final Dataset Preparation

- Merge all cleaned and engineered datasets.
- Generate a final report showing:
    - Number of records before and after cleaning.
    - Number of features created.
    - Missing value summary (before vs. after).
    - Outlier count (before vs. after).

8. Bonus (Optional for Extra Credit)

- Use **Pandas Profiling** or **YData Profiling** to auto-generate an EDA report.
- Save cleaned data as `final_cleaned_dataset.csv`.

📄 Deliverables

1. **Python Notebook** (`main.ipynb`)
2. **Final CSV Output** (`final_cleaned_dataset.csv`)
3. **Summary Report** (Theory + Observations, max 1 page)
4. All of the above three must be within a GitHub repository, including a Readme.md file.